

Javier Herrero Pérez

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About me

I studied Physics because I liked understanding how things work at a fundamental level. That same curiosity pulled me into Data Science and, more recently, into AI agents and LLM systems. I'm currently in my Master's at the University of Valencia, looking for a junior role where I can keep building: ideally on products that use language models or agentic workflows. I have real experience shipping ML pipelines from an internship at AIR Institute, and I'm particularly interested in Generative AI and LLM-based systems.

Technical Skills

Programming: Python, SQL, R, Fortran

Data & ML Libraries: NumPy, Pandas, Scikit-learn

Deep Learning Frameworks: TensorFlow, PyTorch, Keras

MLOps & Tools: Git, AWS, Jupyter Notebook, MongoDB, Cassandra, LaTeX

Languages: Spanish (native), English (B1+: Professional communication and technical proficiency)

Experience

Data Scientist | Internship

Jul 2024 – Aug 2024

AIR Institute (International Centre for Artificial Intelligence Research)

Spain

- **Workplace Risk Classification:** Took a real sensor dataset from scratch and put together a 6-class risk detection pipeline: feature engineering (temporal exposure, exponential decay weighting), SMOTE to handle a 27:1 imbalance, and a Random Forest that hit **100% accuracy and macro F1 = 0.99** on a stratified 80/20 split. Used SHAP to explain the model and show that CO₂ and cumulative exposure time were the dominant drivers. Also trained an XGBoost regressor to predict CO₂ from environmental sensors, hitting **R² = 0.9999 and MAE = 27.74 ppm** (<6% error).
- **Candidate Profiling (NLP):** Built a feature extraction pipeline with Whisper and PySentimiento to pull tone, emotion and linguistic signals from candidate interviews; the features fed a shortlisting classifier and measurably improved its precision and recall.

Projects

Clickbait Detection in News Headlines

Developed an NLP pipeline to detect clickbait using cosine similarity and topic modelling (Python, Scikit-learn, Transformers). Built a LangChain agent that autonomously scrapes news headlines and classifies them as clickbait or not in an end-to-end workflow.

Calorie Estimation from Food Images

Built a deep learning pipeline to estimate calories from images using CNN models (Python, TensorFlow, Hugging Face).

Exploratory Analysis of Labour Market Data

Performed EDA on Spanish labour market data to identify gender inequality patterns (R, tidyverse, ggplot2).

Education

Master's in Data Science

Sept 2025 - Jun 2026 (expected)

University of Valencia

Spain

- GPA: 8.86/10

Bachelor's Degree in Physics

Sept 2021 - Jul 2025

University of Salamanca

Spain

- GPA: 7.25/10